MARYUM AKRAM

AI & Automation Engineer — Agentic Systems · LLM Integration · n8n

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PROFILE

AI engineer who builds autonomous agents and automation systems in Python and n8n and puts them into production, not demos. Independently architected and shipped six end-to-end platforms integrating LLMs (GPT-4o, Gemini, Claude API), RAG retrieval over Supabase/pgvector, voice AI (Retell AI), and third-party APIs (Google Maps, Calendar, Slack, Gmail) across healthcare, e-commerce, home services, and career-tech. Comfortable scoping ambiguous problems solo and defending technical decisions under real constraints — e.g. choosing out-of-core learning to train a classifier on 3.8M+ rows within a 12GB RAM budget — and picking up new tools fast; two projects earned Silver Tier at Google's AI Seekho Hackathon 2026.

TECHNICAL SKILLS

- **Languages & AI:** Python, Prompt Engineering, LLM Integration (GPT-4o, Gemini, Claude API), Claude Code, RAG, pgvector
- **Agentic Systems & Automation:** n8n (multi-step workflow orchestration), Retell AI (voice agents), modular pipeline design
- **Integrations & APIs:** Google Maps, Google Calendar, Slack, Gmail, Google Sheets, REST APIs, webhooks
- **Cloud, Data & Frontend:** Docker, Google Cloud Run, AWS, FastAPI, PostgreSQL, Supabase, Firebase, React, TypeScript

WORK EXPERIENCE

AI Engineering Intern — FlyRank (Backend AI & ML Engineering Tracks) — *6 Jul 2026 – 30 Aug 2026*

- Receipt Extractor (FastAPI, Pydantic, SQLite): schema-validated LLM extraction; returns null + needs_review, never guesses; 8/8 eval. GitHub ↗
- Content Refresh Scoring (ML capstone): client-holdout test of rule baseline vs. LogReg and Random Forest; baseline (P@50 0.90) beat RF (0.76). GitHub ↗
- Embeddable Widget Platform (FastAPI, SQLite): multi-tenant lead capture with proof-of-work bot defense (0% bot success); 47/47 tests. GitHub ↗

PROJECTS

MediLens — AI Hospital Automation Platform — *Personal Project · Flagship* GitHub ↗

Python · n8n · Google Gemini · Supabase (pgvector) · Retell AI · Docker · Google Cloud Run

- Architected and deployed 3 production modules in Python and n8n, containerized with Docker on Google Cloud Run, unifying scheduling, complaints, and admin oversight into one system that runs without manual coordination between hospital departments.
- Built a Retell AI voice agent with Gemini-driven intent classification and RAG-style retrieval over Supabase (pgvector) — patients book, cancel, and reschedule by voice in under 4 minutes with zero staff involvement, replacing a 2-3 person reception desk.
- Designed the automated complaint intake and routing pipeline, turning unstructured caller input into tracked, resolvable tickets.

GharFix — AI Home Services Platform — *AI Seekho 2026, Phase 2 — Silver Tier* GitHub ↗

Python · n8n · Google Gemini · Supabase · React · Google Maps API · Google Calendar API

- Architected an end-to-end marketplace in Python and n8n connecting homeowners with verified providers — matches and books a nearby, vetted provider in under 3 minutes, versus the 1-2 hours it typically takes to call around manually for someone reliable.
- Built proximity-based provider discovery on the Google Maps API and automated job-routing/notification workflows in n8n, eliminating manual dispatch so every booking reaches a provider who's actually nearby and available.

Lumina — AI E-commerce Voice Support Agent — *Personal Project* GitHub ↗

Python · Retell AI · OpenAI · n8n · Google Sheets · Webhooks

- Built an autonomous voice agent resolving e-commerce delivery queries end-to-end, with intent routing for tracking, cancels, returns, refunds.
- Tested across 40-50 live calls: resolved requests in under 2 minutes versus the 5-10 minutes a human agent typically takes, with webhook-based escalation for complex cases.

SkillSync AI — Career Discovery Platform — *Personal Project · Production-Hardened · 9 real users* GitHub ↗

Python · React · Firebase/Firestore · Node.js · Express · GitHub Actions CI

- Built a resume parser and skill-assessment engine matching parsed skills against live job postings with ATS-style scoring, serving 9 real users.
- Hardened the platform against 12 adversarial attack scenarios (identity spoofing, value/resource poisoning) with rate limiting and audit logging, using Claude Code to accelerate the security review and fix cycle; kept CI green with 94/94 tests passing on every merge.

OTHER PROJECTS — CLASSICAL ML (PYTHON)

- **AI vs Human Text Detector —** stylometric + TF-IDF classifier, 99.36% accuracy on 487K essays, deployed on Streamlit. GitHub ↗
- **Consumer Complaint Classifier —** out-of-core pipeline (HashingVectorizer + SGDClassifier) on 3.8M+ rows in a 12GB RAM budget, 85.3% accuracy. GitHub ↗
- **Resume Screener —** two-stage pipeline (TF-IDF + Logistic Regression, Gradient Boosting fit-scoring, R^2 0.814). GitHub ↗

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103, Oct 2026) · Azure AI Fundamentals (AI-900) · AI Seekho 2026 (Google for Developers × Pakistan MoITT) — Silver Tier · EF SET English — B2 Upper Intermediate

EDUCATION

BS Mathematics — Queen Mary College, Lahore · 2014–2018